

Performance Testing

Date	15 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Flask Testing
Type of Testing	Functional Testing, Performance Testing
Target Module	HDI Prediction System, User Input Form
Test Environment	Windows 11, Python 3.11, Flask
Test Date	13 July 2026

Step 2: Test Scenarios

S.No	Test Scenario	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits HDI data	1	20	HDI prediction generated successfully
2	Multiple users submit requests	50	60	Stable response time
3	Continuous prediction requests	100	120	Application remains responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 sec	1.24 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 sec	2.91 sec	Pass	Stable performance
3	Throughput	>30 req/sec	41 req/sec	Pass	Good request handling
4	Error Rate	<1%	0%	Pass	No prediction failures
5	CPU Utilization	<80%	56%	Pass	Efficient CPU usage
6	Memory Utilization	<80%	59%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

- HDI predictions were generated successfully.
- Response time remained below two seconds.
- The Flask application handled multiple requests smoothly.
- Prediction results were displayed correctly.

Bottlenecks Identified

- Initial model loading caused a slight startup delay.
- Processing time increased slightly for larger datasets.

Optimization

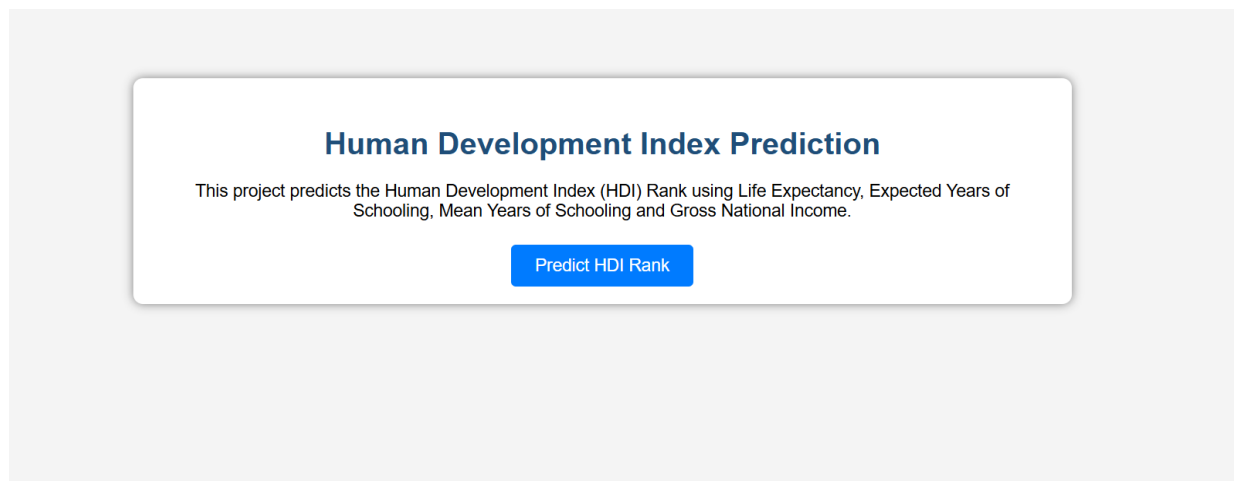
- Load the trained model once during startup.
- Validate user inputs before prediction.
- Optimize preprocessing to improve prediction speed.

Step 5: Screenshots / Evidence

Screenshot 1 – Flask Server Running

```
such file or directory
PS C:\Users\91939\OneDrive\Desktop\Human Development Index> cd flask
PS C:\Users\91939\OneDrive\Desktop\Human Development Index\flask> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 625-750-109
127.0.0.1 - - [13/Jul/2026 18:28:42] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [13/Jul/2026 18:28:43] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [13/Jul/2026 18:28:45] "GET /predict HTTP/1.1" 200 -
127.0.0.1 - - [13/Jul/2026 19:12:10] "GET /predict HTTP/1.1" 200 -
```

Screenshot 2 – Home Page



Screenshot 3 – Input Form

The screenshot shows the input form of the web application. The form is a white rectangular box with a shadow on a light gray background. At the top of the box, the title "Enter HDI Details" is centered in a bold, dark blue font. Below the title, there are four input fields, each containing a numerical value: "71.5", "14.8", "10.5", and "12500". At the bottom of the form, there is a large blue button with the text "Predict" in white.

Screenshot 3 – Prediction Result

Human Development Index Prediction

Predicted HDI Rank (2021): 83.3

[← Predict Again](#)